Review Article

Gut microbiota and acute graft-versus-host disease

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Abstract

Acute graft-versus-host disease (aGVHD) is an important complication which critically impacts the prognosis of patients undergoing allogeneic hematopoietic stem cell transplantation. Increasing evidence suggests that dysbiosis of the gut microbiota plays a key role in aGVHD pathogenesis. The biological process involves compromised intestinal barrier integrity, amplified inflammation driven by the translocation of microbial products like lipopolysaccharide, and finally the dysregulated immune response centralized by T cell activation and differentiation. Meanwhile, certain microbial metabolites such as short-chain fatty acids and secondary bile acids exert protective effects. The clinical relevance of these findings is underscored by studies establishing that specific gut microbial signatures, such as low diversity and single pathogen dominance, independently predict aGVHD morbidity and mortality. From a therapeutic perspective, the microbiome has emerged as an important therapeutic target for aGVHD. Fecal microbiota transplantation has shown significant efficacy in clinical trials for prophylaxis and treatment of aGVHD, providing definitive proof-of-concept for ecological restoration. This review synthesizes these foundational mechanistic insights, from metabolic disruption to host-microbe crosstalk at the mucosal barrier, and details the rapidly advancing clinical landscape of microbiome-targeted diagnostics and therapeutics for aGVHD.

Keywords: Gut microbiota; graft-versus-host disease; fecal microbiota transplantation

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Introduction

The human gastrointestinal (GI) tract harbors a dense and complex ecosystem of microorganisms (including bacteria, archaea, eukaryotes, and viruses), collectively known as the gut microbiota (1). The microbiota functions as a vital metabolic organ critical for digesting dietary fiber, synthesizing essential vitamins, and producing immunomodulatory metabolites such as short-chain fatty acids (SCFAs) (2). In a state of homeostasis, the microbiota plays a fundamental role in training the host immune system and providing colonization resistance against pathogens (3). Dysbiosis of the gut microbiota has also been shown to play a critical role in the pathogenesis of GI, metabolic, cardiovascular, neurological, and musculoskeletal disorders (4-8).

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is a curative therapy for many life-threatening hematologic diseases, but its success is limited by complications. These complications, especially graft-versus-host disease (GVHD), can impair the patient's survcival and quality of life (9-11). Therefore, successful prophylaxis and treatment for GVHD is being actively pursued (12,13). Accumulating evidence has redefined the gut microbiota's role in aGVHD pathogenesis, elevating it from a contributing factor to a cornerstone of the disease process (14). The allo-HSCT procedure itself, particularly pre-transplant conditioning regimens and intensive antibiotic therapies, inflicts a profound "first hit" on this symbiotic ecosystem, leading to severe dysbiosis (15). A retrospective analysis found that the intensity of the conditioning regimen positively correlates with the severity of intestinal dysbiosis (16). During the peri-transplant period, the cytotoxic effects of the conditioning regimen, together with broad-spectrum antibiotics, eliminate beneficial bacteria, particularly those from the class Clostridia which are crucial for gut integrity, while promoting the expansion of opportunistic and pro-inflammatory bacteria, such as *Enterococcus* (17). Relatively, many clinical studies have shown that profound loss of gut microbial diversity after transplantation is an independent predictor for aGVHD morbidity and mortality (18,19). Meawhile, specific microbial signatures, such as the depletion of protective *Clostridia* and the monodominance of pathobionts like *Enterococcus*, have been robustly correlated with poorer clinical outcomes (17).

As a therapeutic intervention, fecal microbiota transplantation (FMT) has already shown efficacy in patients with steroid-refractory aGVHD, and prophylactic use is being investigated (20,21). Beyond this wholesale ecosystem replacement, a spectrum of strategies is being explored, including more targeted microbiome modulation with applications such as prebiotics. This review will synthesize the current understanding of the complex and tripartite crosstalk between the host, gut microbiota and the allogeneic immune system, and will detail the promising clinical progress of these novel microbiome-targeted therapies (*Figure 1*).

Microbial-derived substances and GVHD

Lipopolysaccharide (LPS)

LPS, an endotoxin primarily derived from Gram-negative bacteria, is a key amplifier of the inflammatory cascade in aGVHD following its leakage across the damaged intestinal barrier. Translocated LPS binds to Toll-like receptor 4 (TLR4) on donor-derived antigen-presenting cells, triggering a myeloid differentiation primary response gene 88-dependent signaling pathway that leads to the production of proinflammatory cytokines and the activation of donor-derived T cells, thereby exacerbating intestinal inflammation (22). In a study utilizing a TLR4-deficient murine model, TLR4 deficiency was found to reduce the levels of pro-inflammatory factors such as tumor necrosis factor-α and interleukin-6 (IL-6), inhibit the activation and expansion of donor T cells, and improve post-transplantation survival (23). Additionally, it was also shown that LPS may activate neutrophils, stimulating production of reactive oxygen species and tissue-damaging factors which promote intestinal injury and exacerbate aGVHD (24). In addition to participating in the TLR signaling pathway, LPS was also found to induce the cleavage of Gasdermin D via the activation of Caspase-11. The Gasdermin D cleavage product then forms pores in the cell membrane, mediating the release of multiple proinflammatory factors, such as IL-1α, which ultimately exacerbate aGVHD (25) .

Multiple studies have provided evidence for LPS as a therapeutic target. In murine transplant models, synthetic lipid A analogues (e.g., B975) or high-density lipoprotein infusions lower blood LPS levels, decreasing aGVHD incidence while maintaining the graft-versus-leukemia effect (26,27).

Short-chain fatty acids (SCFAs)

SCFAs, principally butyrate and propionate, are microbial metabolites produced in the colon via bacterial fermentation of dietary fiber by anaerobes such as *Blautia*, *Roseburia*, *Dorea*, and *Faecalibacterium* (28). SCFAs exert potent protective effects by maintaining intestinal homeostasis through several distinct mechanisms. They serve as a primary energy source for intestinal epithelial cells (IECs), promoting tissue regeneration via G-protein-coupled receptor 43 (GPR43), and enhancing epithelial barrier integrity through histone deacetylase inhibition (29-31). In addition, SCFAs also possess profound anti-inflammatory properties. Arpaia *et al.* observed that SCFAs drive the extrathymic differentiation of naïve T cells into anti-inflammatory regulatory T cells in murine models (32). Another study also showed that the abundance of SCFA-producing bacteria is associated with the

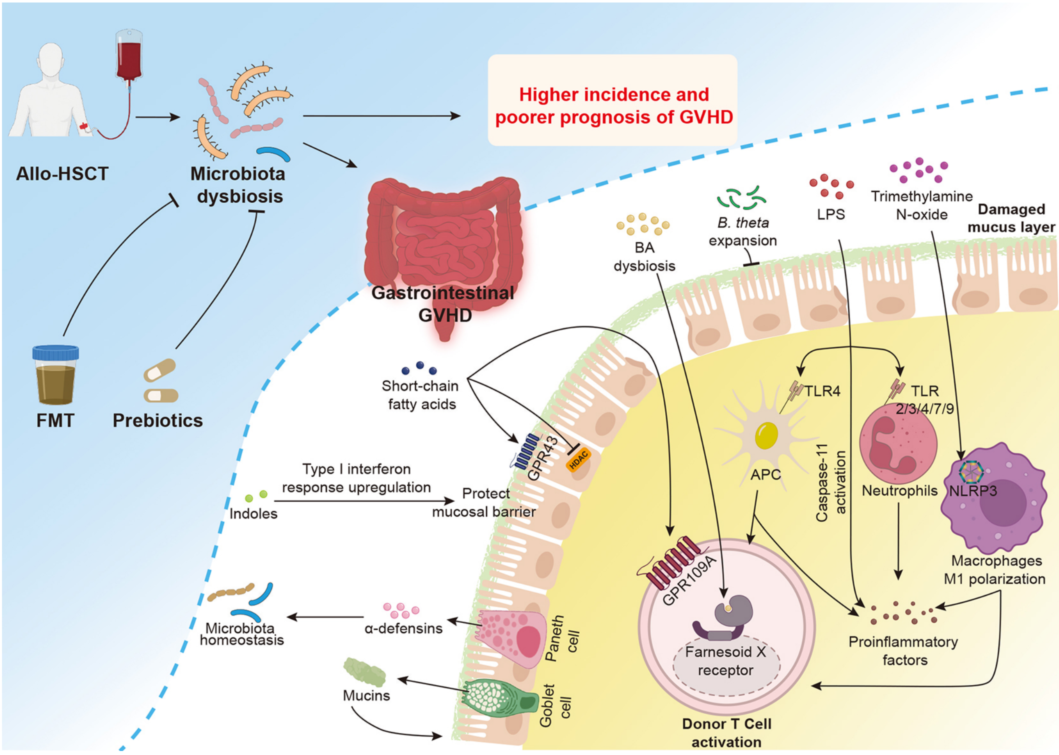

Figure 1 The central role of microbiota dysbiosis in pathogenesis of GI aGVHD following allo-HSCT. During GI aGVHD, microbiota dysbiosis and a compromised mucosal barrier permit the translocation of microbial products that exacerbate the disease. For instance, translocated LPS activates donor T cells and inflammatory pathways via TLR4 and Caspase-11; dysregulated BAs activate T cells via farnesoid X receptor; trimethylamine N-oxide (TMAO) promotes M1 macrophage polarization through the NLRP3 inflammasome; translocated SCFAs may activate donor T cells via GPR109A, and the expansion of *B. theta* can directly degrade the mucus layer. Conversely, protective mechanisms work to maintain gut integrity. Microbial metabolites, including SCFAs (acting via GPR43 and HDAC inhibition) and indole (acting via the Type I interferon response) reinforce the epithelial barrier. Goblet cells secrete protective mucus, and Paneth cells release α-defensins to maintain microbial homeostasis. Therapeutic strategies such as FMT and prebiotics aim to reverse this dysbiosis, restore microbiota homeostasis, and thereby mitigate GVHD. GI, gastrointestinal; aGVHD, acute graft-versus-host disease; allo-HSCT, allogeneic hematopoietic stem cell transplantation; FMT, fecal microbiota transplantation; BA, bile acid; LPS, lipopolysaccharide; GPR, G-protein-coupled receptor; HDAC, histone deacetylase; *B. theta*, *Bacteroides thetaiotaomicron*; TLR, Toll-like receptor; APC, antigen-presenting cell; NLRP3, NOD-like receptor family pyrin domain containing 3 inflammasome.

maintenance of regulatory T cells (33).

Although most studies report protective effects of SCFAs (31-34), their exact role might be more complex and multifaceted. In murine models, the absence of GPR109A on donor T cells was shown to reduce GVHD severity, suggesting this receptor is involved in the expansion and maintenance of pathogenic T cells (35). It is noteworthy that the direct causality between SCFA exposure and GPR109A activation or GVHD exacerbation was not established in this study.

Bile acids (BAs)

The microbiota plays a crucial role in the transformations of BAs through bile salt hydrolases (BSHs) (36). As a major source of BSH, Bacteroides abundance was found to be reduced in aGVHD patients compared to controls in a study that co-analyzed fecal BA metabolomics and 16S rRNA gene sequence. This indicates dysregulation of BA metabolism in aGVHD patients, which may be linked to the progression of aGVHD (37). In a murine model of aGVHD, T cell-mediated inflammation reduced abundance of microbial BSH genes, which led to dysregulation of BA metabolism characterized by reduced levels of unconjugated and microbe-derived BAs. This shift increases bile acid receptor (farnesoid X receptor) activity, leading to more severe aGVHD and adverse clinical outcomes (38).

Indoles

Indole and its derivatives are crucial metabolites produced from dietary tryptophan by various bacterial enzymes (39). Preclinical studies have demonstrated that both endogenous, microbiota-derived indoles and exogenous oral administration can ameliorate GVHD in a murine model. The mechanism includes the activation of the type I interferon signaling pathway, which enhances the integrity of the intestinal mucosal barrier (40).

Trimethylamine N-oxide (TMAO)

Wu *et al.* observed that elevated TMAO activated NOD-like receptor family pyrin domain containing 3 inflammasome by inducing mitochondrial reactive oxygen species, which polarizes macrophages towards a pro-inflammatory M1 phenotype. This environment subsequently promoted the development of pathogenic T helper 1 cells and T helper 17 cells and ultimately GVHD exacerbation (41).

Crosstalk between microbiota and intestinal mucosal barrier

In addition to modulating the immune response of donor-derived cells to intestinal tissues via metabolic byproducts, the gut microbiota interacts with the host's intestinal mucosal barrier, a dynamic complex comprising the mucus layer, the IECs, and resident immune cells.

Mucus layer

The colonic mucus comprises a two-layered system: an outer layer and an inner layer. The latter is normally impervious to bacteria, effectively segregating the microbiota from the epithelial surface. Defects in the inner layer allow pathobionts to contact the epithelium, triggering intestinal inflammation (42). *Bacteroides thetaiotaomicron* (*B. theta*) is a Gram-negative obligate anaerobe with a broad metabolic capacity to utilize diverse substrates, including dietary polysaccharides and host-derived mucin O-glycans, the latter of which is an integral component of the intestinal mucus layer (43). A murine model study revealed that the carbapenem antibiotic meropenem aggravates GVHD by inducing the expansion of *B. theta*, resulting in an increased abundance of mucus-degrading enzymes and degradation of the colonic mucus layer. Notably, oral xylose supplementation improved survival in meropenem-treated mice, suggesting that providing an alternative nutrient source can spare the mucus layer from degradation (44). A subsequent study identified *Bacteroides ovatus* (*B. ovatus*) as a protective species in GVHD, and the reduced frequency of *B. ovatus* in patients is associated with non-response to corticosteroid. In addition, preclinical models demonstrated that administering *B. ovatus* reduces GVHD severity and improves survival. Its protective effect stems from its ability to metabolize dietary polysaccharides, providing an alternative nutrient source that spares the mucus layer from degradation by competitors like *B. theta*, thus mitigating GVHD severity (45).

Intestinal epithelial cells (IECs)

In GVHD, IECs serve as both targets of tissue damage, and potential contributors to disease initiation by expressing major histocompatibility complex class II molecules and presenting antigens directly to donor-derived $CD4^+$ T cells. Microbiota dysbiosis following allo-HSCT was found to promote the expression of major histocompatibility complex class-II on IECs and exacerbate aGVHD (46).

Paneth cells play a crucial homeostatic role by secreting antimicrobial peptides, such as α-defensins, which selectively eliminate non-commensal pathobionts while sparing commensal bacteria, thereby regulating the composition of the intestinal microbiota (47). In GI GVHD patients, analysis of duodenal biopsies revealed that a lower Paneth cell count was significantly associated with more severe GI GVHD, a poor response to therapy, and higher non-relapse mortality (48). In murine models, GVHD depletes Paneth cells and α-defensins, promoting *Escherichia coli* (*E. coli*) overgrowth and worsening GVHD. Oral administration of polymyxin B was shown to inhibit this overgrowth of *E. coli* and ameliorate GVHD (49). Furthermore, in preclinical models, the Wnt agonist R-spondin1 has been shown to stimulate intestinal stem cells to differentiate into Paneth cells, thereby enhancing defensin secretion and helping to restore intestinal microbial homeostasis (50).

Goblet cells secrete mucins, which are the primary structural components of the mucus layer (51). The gut microbiota and host engage in a reciprocal regulatory loop: the microbiota promotes O-glycosylated mucin production and secretion by goblet cells in the proximal colon, while these mucins shape microbial community structure and metabolite production (52). A depletion of intestinal goblet cells is a key feature of GI GVHD. Ara *et al.* discovered that IL-25 maintains the structural and functional integrity of the intestinal mucus layer by protecting goblet cells from GVHD-related immune damage. The protective effect of IL-25 is dependent on the LY6/PLAUR domain containing 8, a protein expressed by IECs which specifically binds to bacterial flagella and inhibits the motility of flagellated bacteria (53,54).

Resident immune cells

Mucosal-associated invariant T (MAIT) cells are critical resident immune cells in the intestinal tissue that participate in local immune regulation by recognizing microbially-derived metabolites (55). In a murine model of aGVHD, host-derived MAIT cells contributed to the maintenance of gut barrier integrity and limited inflammatory responses through the secretion of IL-18 and finally mitigated aGVHD (56). Furthermore, an increased abundance of riboflavin-producing bacteria activate MAIT cells and release gut-protective factors, mitigating gut injury in a dextran sulfate sodium salt model (57).

Predictive and prognostic value of microbiota in aGVHD

The composition of the gut microbiota has emerged as a potent and independent predictor of mortality after allo-HSCT (18). Profound dysbiosis after neutrophil engraftment, marked by a significant loss of microbial diversity, is strongly associated with adverse clinical outcomes (19). Monodominance of pathobionts such as *Enterococcus* or *Streptococcus* and the deficiency of protective butyrate-producing *Clostridia* is strongly associated with a higher incidence of GVHD and lower overall survival (OS) (58).

Pre-transplant gut microbial diversity significantly influences GVHD risk stratification. In murine models, reduced microbial diversity and a depletion of *Clostridia* have been shown to predispose subjects to severe GVHD. This preclinical finding was corroborated by retrospective clinical data showing lower OS in recipients with a high body mass index (>30 kg/m^2) (59). Additionally, lower pre-transplant gut microbial diversity in children strongly predicted reduced OS, highlighting baseline microbial diversity as a robust prognostic indicator (60).

Beyond diversity metrics, more nuanced biomarkers are being identified. In one study analyzing fecal samples from 266 patients undergoing allo-HSCT, researchers found that in addition to the higher abundance of butyrate producers, the higher ratio of strict-to-facultative anaerobes was also associated with the lower risk of GVHD-related mortality post-transplantation (61). Another study found that microbiome signatures associated with protective metabolites were associated with transplant prognosis (62). Longitudinal multi-omic analysis revealed that dynamic microbiota-metabolite trajectories, rather than static biomarkers, more accurately predicted clinical outcomes (63).

Microbiotherapy and aGVHD

Microbiome-targeted interventions for aGVHD are now being rigorously evaluated in clinical trials worldwide. FMT restores intestinal homeostasis by introducing functional fecal material from a healthy donor into a patient's gut via oral capsules, nasogastric tube, or colonoscopy. Clinicians and researchers are actively exploring the value of microbiotherapy as an effective strategy for aGVHD therapy and prophylaxis (*Table 1*).

Treatment

FMT was first introduced into clinical practice as a salvage therapy for patients with severe aGVHD who were refractory to or dependent on corticosteroids. An early retrospective analysis by Kakihana *et al.* reported a 100% overall response rate (ORR) (3 complete and 1 partial response) with a favorable safety profile in four patients with steroid-resistant (n=3) or steroid-dependent (n=1) GI aGVHD (20). These promising signals were subsequently validated in larger prospective cohorts. A recent single-arm prospective trial reported a 28-d ORR of 52.4% in 21 patients with steroid-refractory or -dependent aGVHD receiving single-donor FMT (64).

To overcome the heterogeneity of single-donor FMT and advance toward a standardized biotherapeutic, pooled multi-donor microbiota products have been developed. In a phase 2 trial involving 24 patients with grade III−IV steroid-refractory aGVHD, the pooled product MaaT013 demonstrated a 28-d GI-ORR of 38% (65). More recently, the pivotal phase 3 ARES study (NCT04769895) evaluated MaaT013 as a third-line therapy for patients with GI-aGVHD who had failed both corticosteroids and ruxolitinib. In this exceptionally refractory population (n=66), the GI-ORR reached 62% with an impressive 1-

Table 1 Clinical trials of microbiotherapy for aGVHD prophylaxis and treatment

Microbiotherapy	Clinical trial name and phase	Clinical trial number	Results
Trials on aGVHD second-line or multi-line treatment			
FMT, MaaT013* (From pooled third-party volunteers)	Phase 3	NCT04769895	D 28 GI-ORR rate: 62% 1-year OS: 54% (Median survival not reached)
FMT, Third party	Phase 3	NCT03819803	Results awaited
FMT, MaaT013 (65) (From pooled third-party volunteers)	Phase 2	NCT03359980	HERACLES study D 28 GI-ORR rate: 38% 1-year OS: 25% Compassionate use patients D 28 ORR rate: 49% Steroid and ruxolitinib refractory D 28 GI-ORR: 59% 1-year OS: 72%
FMT, Unmentioned	Phase 2	NCT03812705	Results awaited
FMT, Third party	Phase 1	NCT04285424	Results awaited
FMT, Third party (64)	Phase 1	NCT03214289	D 28 GI-ORR rate: 52.4%
Trials on aGVHD prophylaxis			
FMT, Third party (67)	Phase 2	NCT03678493	FMT-ameliorated intestinal dysbiosis and did not increase infections
FMT, MaaT033 (From pooled third-party volunteers)	Phase 2b	NCT05762211	Results awaited
FMT, Third party	Phase 2	NCT03720392	Results awaited
FMT, Third party (21)	Phase 2	NCT06026371	Single-arm run-in Phase Donor 1-derived (3/6 developed III–IV aGVHD) Donor 2-derived (2/6 developed II aGVHD) Donor 3-derived (1/8 developed II aGVHD)
FMT, Third party	Phase 2	NCT04935684	Results awaited
Prebiotic, galactooligosaccharide	Phase 1/2	NCT04373057	Results awaited
Trials on aGVHD first-line treatment			
FMT, Third party (66)	Phase 1	NCT04139577	Te participants (9 treatment-naïve; 1 steroid-refractory) D 28 LGI-ORR rate: 70%

aGVHD, acute graft-versus-host disease; FMT, fecal microbiota transplantation; GI, gastrointestinal; ORR, overall response rate; OS, overall survival; LGI, lower gastrointestinal. *, Results from the Phase 3 ARES study (NCT04769895) were disclosed in a news release (Maat Pharma, January 8, 2025). Available online: https://www.maatpharma.com/january-8-2025-maat-pharma-announces-positive-topline-results-from-the-pivotal-phase-3-ares-study-evaluating-maat013-in-acute-graft-versus-host-disease/.

year OS of 54% (*Table 1*).

Investigators have begun to explore moving FMT towards first-line therapy for aGVHD. A pioneering open-label, single-arm pilot study (NCT04139577) evaluated the safety and efficacy of combining third-party FMT with systemic corticosteroids as the initial treatment for high-risk, treatment-naïve patients with lower GI aGVHD. In the 10 enrolled patients (9 of whom were treatment-naïve), this combination strategy resulted in a high organ-specific complete response rate of 70% at d 28 (66). This study provides critical, first-in-human evidence for a new paradigm of combining immunosuppression with ecological restoration from day one in high-risk patients and warrants further investigation in larger randomized trials.

Prophylaxis

The success of FMT in treating established aGVHD has given rise to another attractive concept: prophylactic intervention. The core rationale is to preemptively establish a healthy and functional gut ecosystem and prevent the over-activation of the allogeneic immune response. A large, randomized, double-blind phase II trial (NCT03678493) (67) found that prophylactic third-party FMT administered after neutrophil recovery was feasible and safe, which effectively ameliorated dysbiosis by improving microbial diversity. However, this trial did not meet its primary endpoint of reducing the 4-month infection density in patients who underwent allo-HSCT, a finding potentially confounded by the multifactorial nature of infectious complications. Furthermore, a phase II trial (NCT06026371) illustrated the importance of the FMT donor source. In this trial, the GVHD incidence was substantially lower in recipients of stool from one of the three donors, leading to the selection of this "super-donor" for the subsequent randomized phase (21). The pooled-donor oral capsule MaaT033 might address the problem of donor heterogeneity. The PHOEBUS study (NCT05762211), a multi-center randomized phase IIb trial, is currently underway to evaluate if MaaT033 can translate the benefits of microbial restoration into improved 12-month OS and GVHD-free, relapse-free survival.

Beyond FMT, prebiotics are also being explored. Preclinical models have shown that supplementation with galactooligosaccharides can reshape the microbiome and attenuate GVHD (68), leading to the initiation of a multicenter, randomized phase 1/2 trial (NCT04373057). In addition, a preclinical trial validated that a phage-derived antibacterial enzyme can specifically lyse and eliminate intestinal *Enterococcus* faecium, thereby suppressing the onset and progression of aGVHD (69).

Conclusions and perspectives

The gut microbiome is now unequivocally established as a central hub in the pathophysiology of aGVHD, acting as a critical driver, prognostic biomarker, and therapeutic target. Mechanistic insights into how transplant-associated dysbiosis compromises the intestinal barrier and dysregulates the allogeneic T cell response have enabled the use of microbial signatures for patient risk stratification. Microbiotherapy, majorly FMT, has shown safety and promising efficacy in treating and preventing GVHD. More precise next-generation biotherapeutics such as prebiotics and phage-derived enzyme are being investigated. A holistic approach combining low-intensity conditioning and necessary microbiotherapy restoring host-microbiome homeostasis holds the key to lower GVHD incidence and better survival.

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Footnote

Conflicts of Interest: The authors have no conflicts of interest to declare.

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